Hochschule Fulda
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Part 1 – System and Unit Specification (with focus on W5)

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Weather station – Station status monitoring unit(W5)

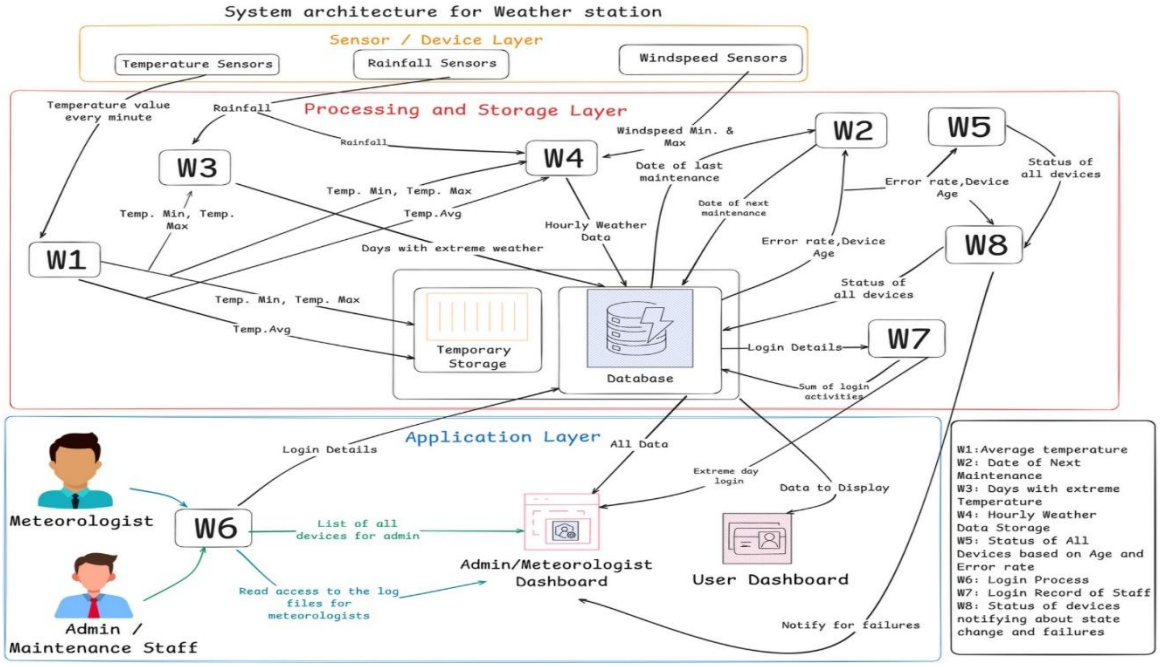
1 System and Unit Specification

1.1 Descriptions of System and Unit

The system collects and analyzes environmental data, including temperature, precipitation, and wind, while monitoring the health of its equipment. It provides accurate weather information and ensures efficient station performance through scheduled maintenance. Meteorologists use it for analysis, and maintenance staff rely on it to keep devices functioning properly.

The Station Status Monitoring unit(W5) evaluates device health and overall station status, coordinating with maintenance to reduce downtime and maintain reliable operation, supporting accurate weather prediction.

1.2 Main System Architecture and Requirements



The architecture is divided into three distinct layers:

1. **Sensor / Device Layer**: Includes temperature, rainfall, and windspeed sensors that capture raw environmental data.
2. **Processing & Storage Layer:** The core of the system, responsible for processing raw inputs into statistics (**W1**), managing database storage (**W4**), and handling maintenance logic (**W2**) and station status checks (**W8**).
3. **Application Layer:** Manages user interaction, including the login process (**W6**), and presents information through the Admin/Meteorologist Dashboard or User Dashboard.

Functional Requirements:

- The system shall collect sensor data hourly and provide meteorologists with access to hourly weather logs.
- The system shall calculate device status (Green/Yellow/Red) based on age and error rates to support proactive maintenance.
- The system shall detect and flag extreme weather conditions (e.g., high temperature or heavy rainfall) and display alerts on the dashboard.

Non-Functional Requirements:

- The system shall maintain at least **99% availability** to ensure no hourly weather data or device status records are lost.
- The system shall protect data integrity by securely storing all logs and encrypting credentials in the database.

1.3 Requirements for Station status monitoring unit(W5)

Functional Requirements:

- The System shall collect and aggregate device health data, including age, error rates, and performance metrics, to monitor overall station health.
- The System shall calculate and update each device's status in real-time as Green, Yellow, or Red to reflect current performance.
- Trigger alerts to Maintenance Scheduling (W2) unit when device performance falls.
- Log all device status changes in the database for future analysis.

Non-Functional Requirements:

- The System shall ensure fast processing to maintain continuous monitoring.
- The System shall maintain high reliability and uptime.
- The System shall calculate device status accurately with minimal error.

Interfaces of unit 5 to other components in the system:

- **Interface with W2 (Maintenance Scheduling):** Unit W8 sends device health and performance data to W2 for proactive maintenance scheduling.
- **Interface with W8 (Station Status Monitoring):** Unit W8 shares device performance data to keep the overall station status updated.
- **Interface with Database (System Database):** Unit W8 reads device metadata and writes logs of status changes for record-keeping and analysis.
- **Interface with Display (Dashboard):** Unit W8 sends device statuses to the dashboard for real-time user monitoring.